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MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Section 5.2

The Characteristic Equation

Eigenvalues and Eigenvectors

An eigenvector of an $n \times n$ matrix A is a nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ .

A scalar λ is called an eigenvalue of A if there is a nontrivial solution $\mathbf{x}$ of $A\mathbf{x} = \lambda\mathbf{x}$. Such an $\mathbf{x}$ is called an eigenvector of A corresponding to λ

Finding Eigenvalues

The eigenvalue-eigenvector equation $A\mathbf{x} = \lambda\mathbf{x}$ is equivalent to $(A - \lambda I)\mathbf{x} = \mathbf{0}$.

For this equation to have a nontrivial solution, $(A - \lambda I)$ must *not* be invertible.

Thus, we seek values of λ so that

$$\det(A - \lambda I) = 0$$

Example 1

Find the eigenvalues of $A = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$.

The Characteristic Equation

A scalar λ is an eigenvalue of an $n \times n$ matrix A if and only if λ satisfies the characteristic equation

$$\det(A - \lambda I) = 0$$

$\det(A - \lambda I)$ is a polynomial of degree n , and it is called the characteristic polynomial.

The multiplicity of an eigenvalue is its multiplicity as a root of the characteristic equation.

Example 2

Find the characteristic polynomial and the eigenvalues of

$$A = \begin{bmatrix} 3 & -4 \\ 4 & 8 \end{bmatrix}$$

Example 3

Find the characteristic polynomial and the eigenvalues of

$$A = \begin{bmatrix} 4 & 0 & 0 \\ 5 & 3 & 2 \\ -2 & 0 & 2 \end{bmatrix}$$

Example 4

Find the eigenvalues of A and their multiplicities. Then, determine the value(s) of h , if any, for which the eigenspace for $\lambda = 5$ is two-dimensional.

$$A = \begin{bmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & h & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
